

VECTOR DATABASE

IVF indexing + RaBitQ quantization

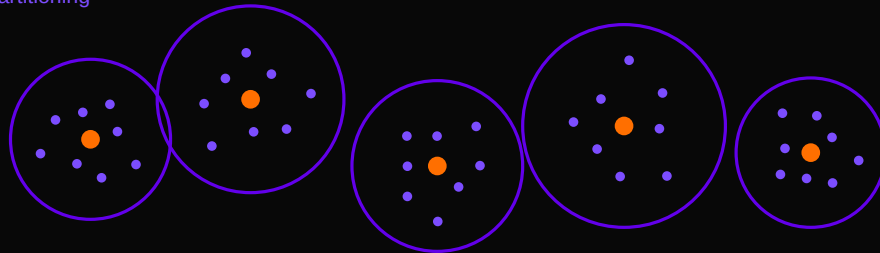
INPUT

raw vector embeddings



IVF

cluster partitioning



nprobe → subset search

RaBitQ

binary compression



32-bit



1-bit

hamming distance → fast approximation

SEARCH

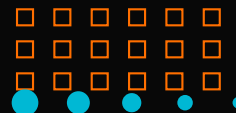
two-stage pipeline

coarse

hamming filter

rerank

exact L2 computation



OUTPUT

top-k results



32:1 compression

~95% recall

√2 exact distances

inverted file | k-means clustering | randomized orthogonal transform | hamming approximation
binary quantization | L2 distance | exact reranking | vector database